Worksheet 2-5 & 2-6 A
Power Rule and Basic Derivative Rules #1
AP Calculus Name_____

Directions: For the following functions, find the derivative.

1. $f(x) = 3x^4 - x$

2. $f(x) = 3 - 4x$

3. $f(x) = e^2$

4. $f(x) = x^5 - 2x^2 + \frac{2}{3}$

5. $f(x) = \frac{4}{3}x^6 + x$

6. $f(x) = \frac{x}{3} - \frac{x^2}{4} + \frac{5}{6}$

7. $f(x) = \frac{1}{x} - \frac{4}{x^2} + x^3$

8. $f(x) = \frac{1}{\pi^3} + \frac{2x}{5} - \frac{5}{x^4} + 7$

9. $f(x) = \frac{2x^3 - 4x + 5}{7}$

10. $f(x) = (3x - 4)(2x + 1)$

11. $h(x) = \sqrt{x}$

12. $h(x) = 3\sqrt{x} - \sqrt[3]{x}$

13. $h(x) = \sqrt{5} - 2\sqrt[5]{x} + 4x^3$

14. $h(x) = \frac{\sqrt[2]{x}}{4} - 4x + \frac{3}{x}$

15. $h(x) = \frac{3}{\sqrt{x}}$

16. $h(x) = \frac{2}{\sqrt[3]{x}} - \sqrt{\pi}$

17. $h(x) = \sqrt{x^5} + 3$

18. $h(x) = 3\sqrt{x} - 6\sqrt[3]{x}$

19. $h(x) = \dfrac{4\sqrt{x}}{3}$

20. $h(x) = \dfrac{7}{5\sqrt{x}} - \dfrac{\sqrt[3]{x}}{2}$

21. $g(x) = x^{5/3} - x^{\frac{2}{3}}$

22. $g(x) = \dfrac{x^2 + 4x + 3}{\sqrt{x}}$

For 23 – 24, write the equation of the line tangent to g(x) at x = −3.

23. $g(x) = 3x^2 - 4x$

24. $g(x) = \dfrac{6}{x} - \dfrac{9}{x^2}$

25. Find the points on the curve $y = 2x^3 + 3x^2 - 12x + 1$ where the tangent is horizontal.